Beyond Accuracy: What Data Quality Means to Data Consumers

RICHARD Y. WANG AND DIANE M. STRONG

RICHARD Y. WANG is Associate Professor of Information Technologies (IT) and Co-Director for Total Data Quality Management (TDQM) at the MIT Sloan School of Management, where he received a Ph.D. degree with an IT concentration. He is a major proponent of data quality research, with more than twenty papers written to develop a set of concepts, models, and methods for this field. Professor Wang received more than one million dollars of research grants from both the public and private sector. His work on data quality was applied, by the Navy, to the Naval Command, Control, Communication, Computers, and Intelligence (C^4I) information architecture. He presented the state-of-the-art of data quality research and practice in the Chief Information Officer (CIO) conference in 1993, and spoke on "Data Quality in the Information Highways" at the Enterprise '93 conference. Dr. Wang organized the first Workshop on Information Technologies and Systems (WITS) in 1991. At WITS–94 he was elected chairman of the Executive Steering Committee. He also chaired or participated in the data quality panel at WITS and at the International Conference on Information Systems. Dr. Wang is the editor of *Information Technologies. Trends and Perspectives.*

DIANE M. STRONG is Assistant Professor of Management at Worcester Polytechnic Institute. She received her Ph.D. in information systems from Carnegie Mellon University, an M.S. in computer science from the New Jersey Institute of Technology, and a B.S. in computer science from the University of South Dakota. Dr. Strong's research centers on data and information quality and software quality. Her publications have appeared in *ACM Transactions on Information Systems, MIS Quarterly, Decision Support Systems*, and other leading journals.

ABSTRACT: Poor data quality (DQ) can have substantial social and economic impacts. Although firms are improving data quality with practical approaches and tools, their improvement efforts tend to focus narrowly on accuracy. We believe that data consumers have a much broader data quality conceptualization than IS professionals realize. The purpose of this paper is to develop a framework that captures the aspects of data quality that are important to data consumers.

A two-stage survey and a two-phase sorting study were conducted to develop a hierarchical framework for organizing data quality dimensions. This framework

Acknowledgments: Research conducted herein has been supported, in part, by MIT's Total Data Quality Management (TDQM) Research Program and MIT's International Financial Services Research Center (IFSRC). The authors wish to thank Lisa Guarascio for her field work and factor analysis, Professors Donald Ballou, Izak Benbasat, and Stuart Madnick for providing insightful comments and encouragement, Professors France LeClerc and Paul Berger for their input on the design and execution of this research, and Daphne Png for conducting the confirmatory experiment and summarizing its results.

Journal of Management Information Systems / Spring 1996, Vol 12, No 4, pp. 5–34
Copyright © 1996 M E Sharpe, Inc

Copyright © 2001 All Rights Reserved

captures dimensions of data quality that are important to data consumers. *Intrinsic DQ* denotes that data have quality in their own right. *Contextual DQ* highlights the requirement that data quality must be considered within the context of the task at hand. *Representational DQ* and *accessibility DQ* emphasize the importance of the role of systems. These findings are consistent with our understanding that high-quality data should be intrinsically good, contextually appropriate for the task, clearly represented, and accessible to the data consumer.

Our framework has been used effectively in industry and government. Using this framework, IS managers were able to better understand and meet their data consumers' data quality needs. The salient feature of this research study is that quality attributes of data are collected from data consumers instead of being defined theoretically or based on researchers' experience. Although exploratory, this research provides a basis for future studies that measure data quality along the dimensions of this framework.

KEY WORDS AND PHRASES: data administration, data quality, database systems.

Introduction

MANY DATABASES ARE NOT ERROR-FREE, AND SOME CONTAIN a surprisingly large number of errors [3, 4, 5, 21, 28, 34, 38, 40]. A recent industry report, for example, notes that more than 60 percent of the surveyed firms (500 medium-size corporations with annual sales of more than $20 million) have problems with data quality.[1] Data quality problems, however, go beyond accuracy to include other aspects such as completeness and accessibility. A big New York bank found that the data in its credit-risk management database were only 60 percent complete, necessitating double-checking by anyone using it.[2] A major manufacturing company found that it could not access all sales data for a single customer because many different customer numbers were assigned to represent the same customer. In short, poor data quality can have substantial social and economic impacts.

To improve data quality, we need to understand what data quality means to data consumers (those who use data). The purpose of this research, therefore, is to develop a framework that captures the aspects of data quality that are important to data consumers.

Related Research

The concept of "fitness for use" is now widely adopted in the quality literature. It emphasizes the importance of taking a consumer viewpoint of quality because ultimately it is the consumer who will judge whether or not a product is fit for use [13, 15, 22, 23]. In this research, we also take the consumer viewpoint of "fitness for use" in conceptualizing the underlying aspects of data quality. Following this general quality literature, we define "data quality" as *data that are fit for use by data consumers*. In addition, we define a "data quality dimension" as a set of data quality attributes that represent a single aspect or construct of data quality.

Copyright © 2001 All Rights Reserved

Three approaches are used in the literature to study data quality: (1) an intuitive, (2) a theoretical, and (3) an empirical approach. The intuitive approach is taken when the selection of data quality attributes for any particular study is based on the researchers' experience or intuitive understanding about what attributes are "important." Most data quality studies fall into this category. The cumulative effect of these studies is a small set of data quality attributes that are commonly selected. For example, many data quality studies include accuracy as either the only or one of several key dimensions [4, 28, 32]. In the accounting and auditing literature, reliability is a key attribute used in studying data quality [7, 11, 21, 24, 25, 49].

In the information systems literature, *information quality* and *user satisfaction* are two major dimensions for evaluating the success of information systems [12]. These two dimensions generally include some data quality attributes, such as *accuracy, timeliness, precision, reliability, currency, completeness*, and *relevancy* [2, 20, 27]. Other attributes such as *accessibility* and *interpretability* are also used in the data quality literature [44, 45].

Most of these studies identify multiple dimensions of data quality. Furthermore, although a hierarchical view of data quality is less common, it is reported in several studies [27, 34, 44]. None of these studies, however, empirically collects data quality attributes from data consumers.

A theoretical approach to data quality focuses on how data may become deficient during the data manufacturing process. Although theoretical approaches are often recommended, research offers few examples. One such study uses an ontological approach in which attributes of data quality are derived on the basis of data deficiencies, which are defined as the inconsistencies between the view of a real-world system that can be inferred from a representing information system and the view that can be obtained by directly observing the real-world system [42].

The advantage of using an intuitive approach is that each study can select the attributes most relevant to the particular goals of that study. The advantage of a theoretical approach is the potential to provide a comprehensive set of data quality attributes that are intrinsic to a data product. The problem with both of these approaches is that they focus on the product in terms of development characteristics instead of use characteristics. They fail to capture the voice of the consumer. Evaluations of theoretical approaches to defining product attributes as a basis for improving quality find that they are not an adequate basis for improving quality and are significantly worse than empirical approaches.

To capture the data quality attributes that are important to data consumers, we take an empirical approach. An empirical approach to data quality analyzes data collected from data consumers to determine the characteristics they use to assess whether data are fit for use in their tasks. Therefore, these characteristics cannot be theoretically determined or intuitively selected by researchers. The advantage of an empirical approach is that it captures the voice of customers. Furthermore, it may reveal characteristics that researchers have not considered as part of data quality. The disadvantage is that the correctness or completeness of the results cannot be proven via fundamental principles.

Copyright © 2001 All Rights Reserved

Research Approach

We follow the methods developed in marketing research for determining the quality characteristics of products. Our approach implicitly assumes that data can be treated as a product. It is an appropriate approach because an information system can be viewed as a data manufacturing system acting on raw data input to produce output data or data products [1, 6, 16, 19, 35, 43, 46]. While most data consumers are not purchasing data, they are choosing to use or not to use data in a variety of tasks.

Approaches for assessing product quality attributes that are important to consumers are well established in the marketing discipline [8, 26]. Three tasks are suggested in identifying quality attributes of a product: (1) identifying consumer needs, (2) identifying the hierarchical structure of consumer needs, and (3) measuring the importance of each consumer need [17, 18].

Following the marketing literature, this research identifies the attributes of data quality that are important to data consumers. We first collect data quality attributes from data consumers, and then collect importance ratings for these attributes and structure them into a hierarchical representation of data consumers' data quality needs. Our goal is to develop a comprehensive, hierarchical framework of data quality attributes that are important to data consumers.

Some researchers may doubt the validity of asking consumers about important quality attributes because of the well-known difficulties with evaluating users' satisfaction with information systems [30]. Importance ratings and user satisfaction, however, are two different constructs. Griffin and Hauser [17], for example, demonstrate that determining attributes of importance to consumers, collecting importance ratings of these attributes, and measuring attribute values are valid characterizations of consumers' actions such as purchasing the product, but satisfaction ratings of these attributes are uncorrelated with consumer actions.

Research Method

We first developed two surveys that were used to collect data from data consumers (referred to as the two-stage survey later). The first survey produced a list of possible *data quality attributes*, attributes that came to mind when the data consumer thought about data quality. The second survey assessed the importance of these possible data quality attributes to data consumers. The importance ratings from the second survey were used in an exploratory factor analysis to yield an intermediate set of *data quality dimensions* that were important to data consumers.

Because the detailed surveys produced a comprehensive set of data quality attributes for input to factor analysis, a broad spectrum of intermediate data quality dimensions were revealed. We conducted a follow-up empirical study to group these intermediate data quality dimensions for the following reasons. First, it is probably not critical for evaluation purposes to consider so many quality dimensions [27]. Second, although these dimensions can be ranked by the importance ratings, the highest ranking dimensions may not capture the essential aspects of data quality. Third, the interme-

Copyright © 2001 All Rights Reserved

diate dimensions seem to form several families of factors. Grouping these intermediate data quality dimensions into families of factors is consistent with research in the marketing discipline. For example, Deshpande [14] grouped *participation in decision making* and *hierarchy of authority* together as a family, named *centralization* factors, and *job codification* and *job specificity* as a family, named *formalization* factors.

In grouping these intermediate dimensions into families, we used a preliminary conceptual framework developed from our experience with data consumers. This conceptual framework consisted of four "ideal" or target categories. Our intent was to evaluate the extent to which the intermediate dimensions matched these categories. Thus, our follow-up study moved beyond the purely exploratory nature of the two-stage survey to a more confirmatory study.

This follow-up study consisted of two phases (referred to later as the two-phase study). For the first phase, subjects were instructed to sort these dimensions into categories, and then label the categories. For the second phase, a different set of subjects was instructed to sort these dimensions into the categories revealed from the first phase to confirm these findings.

The key result of this research is a comprehensive framework of data quality from data consumers' perspectives. Such a framework serves as a foundation for improving the data quality dimensions that are important to data consumers. Our analysis is oriented toward the characteristics of the quality of data in use, in addition to the characteristics of the quality of data in production and storage; therefore, it extends the concept of data quality beyond the traditional development view. Our results have been used effectively in industry and government. Several Fortune 100 companies and the U.S. Navy [33] have used our framework to identify potential areas of data deficiencies, operationalize the measurements of these data deficiencies, and improve data quality along these measures.

Preliminary Conceptual Framework

Based on the limited relevant literature, the concept of fitness for use from the quality literature, and our experiences with data consumers, we propose a preliminary conceptual framework for data quality that includes the following aspects:

- The data must be *accessible* to the data consumer. For example, the consumer knows how to retrieve the data.
- The consumer must be able to *interpret* the data. For example, the data are not represented in a foreign language.
- The data must be *relevant* to the consumer. For example, data are relevant and timely for use by the data consumer in the decision-making process.
- The consumer must find the data *accurate*. For example, the data are correct, objective and come from reputable sources.

Although we hypothesize that any data quality framework that captures data consumers' perspectives of data quality will include the above aspects, we do not bias our initial data collection in the direction of our conceptualization. To be unbiased,

Copyright © 2001 All Rights Reserved

we start with an exploratory approach that includes not only the attributes in our preliminary framework, but also the attributes mentioned in the literature. For example, our first questionnaire starts with some attributes (timeliness and availability) that are not part of this preliminary framework.

The Two-Stage Survey

THE PURPOSE OF THIS TWO-STAGE SURVEY IS TO IDENTIFY data quality dimensions perceived by data consumers. In the following, we summarize the method and key results of these surveys. The reader is referred to [47] for more detailed results.

Method

The method for the two-stage survey was as follows: For stage 1, we conducted a survey to generate a list of data quality attributes that capture data consumers' perspectives of data quality. For stage 2, we conducted a survey to collect data on the importance of each of these attributes to data consumers, and then performed an exploratory factor analysis on the importance data to develop an intermediate set of data quality dimensions.[3]

The First Survey

The purpose of the first survey was to generate an extensive list of potential data quality attributes. Since the data quality dimensions resulting from factor analysis depend, to a large extent, on the list of attributes generated from the first survey, we decided that (1) the subjects should be data consumers who have used data to make decisions in diverse contexts within organizations, and (2) we should be able to probe and question the subjects in order to fully understand their answers.

Subjects: Two pools of subjects were selected. The first consisted of 25 data consumers currently working in industry. The second was M.B.A. students at a large U.S. university. We selected 112 students who had work experience as data consumers. The average age of these students was over 30.

Survey Instrument: The survey instrument (see appendix A) included two sections for eliciting data quality attributes. The first section elicited respondents' first reaction to data quality by asking them to list those attributes that first came to mind when they thought of data quality (beyond the common attributes of timeliness, accuracy, availability, and interpretability). The second section provided further cues by listing 32 attributes beyond the four common ones to "spark" any additional attributes. These 32 attributes were obtained from data quality literature and discussions among data quality researchers.

Procedure: For the selected M.B.A. students, the survey was self-administered. For the subjects working in industry, the administration of the survey was followed by a discussion of the meanings of the attributes the subjects generated.

Results: This process resulted in 179 attributes, as shown in figure 1.

Copyright © 2001 All Rights Reserved

Ability to be Joined With	Ability to Download	Ability to Identify Errors	Ability to Upload
Acceptability	Access by Competition	Accessibility	Accuracy
Adaptability	Adequate Detail	Adequate Volume	Aestheticism
Age	Aggregatability	Alterability	Amount of Data
Auditable	Authority	Availability	Believability
Breadth of Data	Brevity	Certified Data	Clarity
Clarity of Origin	Clear Data Responsibility	Compactness	Compatibility
Competitive Edge	Completeness	Comprehensiveness	Compressibility
Concise	Conciseness	Confidentiality	Conformity
Consistency	Content	Context	Continuity
Convenience	Correctness	Corruption	Cost
Cost of Accuracy	Cost of Collection	Creativity	Critical
Current	Customizability	Data Hierarchy	Data Improves Efficiency
Data Overload	Definability	Dependability	Depth of Data
Detail	Detailed Source	Dispersed	Distinguishable Updated Files
Dynamic	Ease of Access	Ease of Comparison	Ease of Correlation
Ease of Data Exchange	Ease of Maintenance	Ease of Retrieval	Ease of Understanding
Ease of Update	Ease of Use	Easy to Change	Easy to Question
Efficiency	Endurance	Enlightening	Ergonomic
Error-Free	Expandability	Expense	Extendibility
Extensibility	Extent	Finalization	Flawlessness
Flexibility	Form of Presentation	Format	Integrity
Friendliness	Generality	Habit	Historical Compatibility
Importance	Inconsistencies	Integration	Integrity
Interactive	Interesting	Level of Abstraction	Level of Standardization
Localized	Logically Connected	Manageability	Manipulable
Measurable	Medium	Meets Requirements	Minimality
Modularity	Narrowly Defined	No lost information	Normality
Novelty	Objectivity	Optimality	Orderliness
Origin	Parsimony	Partitionability	Past Experience
Pedigree	Personalized	Pertinent	Portability
Preciseness	Precision	Proprietary Nature	Purpose
Quantity	Rationality	Redundancy	Regularity of Format
Relevance	Reliability	Repetitive	Reproducibility
Reputation	Resolution of Graphics	Responsibility	Retrievability
Revealing	Reviewability	Rigidity	Robustness
Scope of Info	Secrecy	Security	Self-Correcting
Semantic Interpretation	Semantics	Size	Source
Specificity	Speed	Stability	Storage
Synchronization	Time-independence	Timeliness	Traceable
Translatable	Transportability	Unambiguity	Unbiased
Understandable	Uniqueness	Unorganized	Up-to-Date
Usable	Usefulness	User Friendly	Valid
Value	Variability	Variety	Verifiable
Volatility	Well-Documented	Well-Presented	

Figure 1. Data Quality Attributes Generated from the First Survey

Copyright © 2001 All Rights Reserved

The Second Survey

The purpose of the second survey was to collect data about the importance of quality attributes as perceived by data consumers. The results of the second survey were ratings of the importance of the data quality attributes. These importance ratings were the input for an exploratory factor analysis to consolidate these attributes into a set of data quality dimensions.

Subjects: Since we needed a sample consisting of a wide range of data consumers with different perspectives, we selected the alumni of the M.B.A. program of a large university who reside in the United States. These alumni consisted of individuals in a variety of industries, departments, and management levels who regularly used data to make decisions, thus satisfying the requirement for data consumers with diverse perspectives. From over 3,200 alumni, we randomly selected 1,500 subjects.

Survey Instrument: The list of attributes shown in figure 1 was used to develop the second survey questionnaire (see appendix B). The questionnaire asked the respondent to rate the importance of each data quality attribute for their data on a scale from 1 to 9, where 1 was extremely important and 9 not important. The questionnaire was divided into four sections, depending on the appropriate wording of the attributes, for example, as stand-alone adjectives or as complete sentences. Since we did not include definitions with the attributes, it is possible that data consumers responding to the surveys could interpret the meanings of the attributes differently. Attributes that are not important or that are not consistently interpreted across data consumers will not show up as significant in the factor analysis.

A pretest of the questionnaire was administered to fifteen respondents: five industry executives, six professionals, two professors, and two M.B.A. students. Minor changes were made in the format of the survey as a result of the pretest. Based on the results from the pretest, the final second survey questionnaire included 118 data quality attributes (i.e., 118 items for factor analysis) to be rated for their importance, as shown in appendix B.

Procedure: This survey was mailed along with a cover letter explaining the nature of the study, the time to complete the survey (less than twenty minutes), and its criticality. Most of the alumni addresses were home addresses. To assure a successful survey, we sent the survey questionnaires via first-class mail. We gave respondents a six-week cut-off period to respond to the survey.

Response Rate: Of the 1,500 surveys mailed, sixteen were returned as undeliverable. Of the remaining 1,484, 355 viable surveys (an effective response rate of 24 percent) were returned by the six-week deadline.[4]

Missing Responses: While none of the 118 attributes (items) had 355 responses, none had fewer than 329 responses. There did not appear to be any significant pattern to the missing responses.

Results: Descriptive statistics of the 118 items (attributes) are presented in appendix C. Most of the 118 items had a full range of values from 1 to 9, where 1 means extremely important and 9 not important. The exceptions were accuracy, reliability, level of detail, and easy identification of errors. Accuracy and reliability had the

Copyright © 2001 All Rights Reserved

smallest range, with values ranging from 1 to 7; level of detail and easy identification of errors ranged from 1 to 8. Ninety-nine of the 118 items (85 percent) had means less than or equal to 5; that is, most of the items surveyed were considered to be important data quality attributes. Two items—*accuracy* and *correct*—had means less than 2 and thus were overall the most important data quality attributes, with means of 1.771 and 1.816, respectively. An exploratory factor analysis of the importance ratings produced twenty dimensions, as shown in Table 1. As mentioned earlier, more detailed results can be found in [47].

Factor Interpretation: Factor analysis is appropriate for this study because its primary application is to uncover an underlying data structure [10, 37]. An alternative method would be to ask subjects to group the attributes into common dimensions, as we did in the second phase of this research. Grouping tasks provide more assurance that the factors are interpretable. While both factor analysis and grouping tasks are used in the literature to uncover dimensions, grouping tasks become impractical when the number of items increases. Furthermore, factor analysis can uncover dimensions that are not obvious to researchers.

A potential disadvantage of factor analysis in this research is that attributes with nothing in common could load on the same factor because they have the same importance ratings. This would lead to problems in interpreting the factors. Importance ratings collected from a sample of 355 data consumers with diverse backgrounds, however, will be similar only if data consumers perceive these items consistently. If these items form different constructs, data consumers will rate their importance differently, resulting in different factors. Furthermore, the twenty dimensions show face validity because it was easy for us to name these factors.

Factor Stability: Since the number of survey responses relative to the number of attributes is lower than recommended, it is possible that the factor structure is not sufficiently stable. To test factor stability, we reran the analysis using two different approaches. First, in a series of factor analysis runs, we varied the number of factors to test whether the attributes loading on those factors changed as the number of factors changed. Second, we ran the factor analysis using as input only those attributes that actually loaded on the twenty dimensions to test whether the insignificant attributes affected the results.

In the first approach, our analysis of factor stability found that fourteen out of the twenty dimensions were stable across the series of runs. That is, the same dimensions consisting of the same attributes emerged (see Table 1). Two dimensions, 15 (ease of operation) and 20 (flexibility), were stable in terms of the attributes loading on them, but were combined into a single dimension in runs fixed at fewer dimensions. Four dimensions, 5, 9, 16, and 19, which are all single-attribute factors, have less than desirable stability. Specifically, in some runs, these dimensions either were not significant or they were combined with another single-attribute dimension.

In the second approach, the factor analysis used the 71 attributes shown in Table 1, which meets responses-to-attribute ratio recommendations. The second approach produced the same results as the first. Thus, we conclude that these dimensions are stable with the caveat that additional research is needed to verify most of the single-attribute dimensions.

Copyright © 2001 All Rights Reserved

Table 1. Description of the Dimensions

Dim.	Name of dimension (attribute list)	Mean	S.D.	C.I.	Cronbach α
1	Believability (believable)	2.71	0.10	2.51–2.91	N/A
2	Value-added (data give you a competitive edge, data add value to your operations)	2.83	0.09	2.65–3.01	0.70
3	Relevancy (applicable, relevant, interesting, usable)	2.95	0.06	2.82–3.08	0.69
4	Accuracy (data are certified error-free, accurate, correct, flawless, reliable, errors can be easily identified, the integrity of the data, precise)	3.05	0.10	2.86–3.24	0.87
5	Interpretability (interpretable)	3 20	0.09	3.03–3.37	N/A
6	Ease of understanding (easily understood, clear, readable)	3.22	0.07	3.07–3.37	0.79
7	Accessibility (accessible, retrievable, speed of access, available, up-to-date)	3.47	0.08	3.32–3.62	0.81
8	Objectivity (unbiased, objective)	3.58	0.09	3.40–3.76	0.73
9	Timeliness (age of data)	3.64	0 11	3 43–3.85	N/A
10	Completeness (breadth, depth, and scope of information contained in the data)	3.88	0.09	3.74–4.06	0.98
11	Traceability (well-documented, easily traced, verifiable)	3.97	0.09	3.79–4.14	0.79
12	Reputation (reputation of the data source, reputation of the data)	4 04	0.10	3.83–4.25	0.87
13	Representational consistency (data are continuously presented in same format, consistently represented, consistently formatted, data are compatible with previous data)	4.22	0.09	4.04–4.39	0.84
14	Cost-effectiveness (cost of data accuracy, cost of data collection, cost-effective)	4.25	0.10	4.05–4.44	0.85

Copyright © 2001 All Rights Reserved

Table 1. *Continued*

Dim.	Name of dimension (attribute list)	Mean	S.D.	C.I.	Cronbach α
15	Ease of operation (easily joined, easily changed, easily updated, easily downloaded/uploaded, data can be used for multiple purposes, manipulable, easily aggregated, easily reproduced, data can be easily integrated, easily customized)	4.28	0.08	4.13–4.44	0.90
16	Variety of data and data sources (you have a variety of data and data sources)	4.71	0.12	4.48–4.95	N/A
17	Concise (well-presented, concise, compactly represented, well-organized, aesthetically pleasing, form of presentation, well-formatted, format of the data)	4.75	0.08	4.59–4.92	0.92
18	Access security (data cannot be accessed by competitors, data are of a proprietary nature, access to data can be restricted, secure)	4.92	0.11	4.70–5.14	0.84
19	Appropriate amount of data (the amount of data)	5.01	0.11	4.79–5.23	N/A
20	Flexibility (adaptable, flexible, extendable, expandable)	5.34	0.09	5.17–5.51	0.88

A dimension mean was computed as the average of the responses to all of the items with a loading of 0.5 or greater on the dimension. For example, the dimension ease of understanding consisted of the three items: *easily understood, readable*, and *clear*. The mean importance for *ease of understanding* was the average of the importance ratings for easily understood, readable, and clear. (See Table 1 for the means, standard deviations, and confidence intervals.)

Cronbach's alpha, a measure of construct reliability, was computed for each dimension to assess the reliability of the set of items forming that dimension. As shown in the rightmost column of Table 1, these alpha coefficients ranged from 0.69 to 0.98. As a rule, alphas of 0.70 or above represent satisfactory reliability of the set of items measuring the construct (dimension). Thus, the items measuring our dimensions are sufficiently reliable.

Copyright © 2001 All Rights Reserved

Table 2. Four Target Categories for the 20 Dimensions

Target category	Dimension	Adjustment
Accuracy of data	Believability	None
	Accuracy	None
	Objectivity	None
	Completeness	Moved to category 2
	Traceability	Eliminated
	Reputation	None
	Variety of Data Sources	Eliminated
Relevancy of data	Value-added	None
	Relevancy	None
	Timeliness	None
	Ease of operation	Eliminated
	Appropriate amount of data	None
	Flexibility	Eliminated
Representation of data	Interpretability	None
	Ease of understanding	None
	Representational consistency	None
	Concise representation	None
Accessibility of data	Accessibility	None
	Cost-effectiveness	Eliminated
	Access security	None

Note: A target category is a hypothesized category based on our preliminary conceptual framework.

The Two-Phase Sorting Study

TWENTY DIMENSIONS WERE TOO MANY FOR PRACTICAL EVALUATION PURPOSES. In addition, although these dimensions were ranked by the importance ratings, the highest-ranking dimensions might not capture the essential aspects of data quality. Finally, a grouping of these dimensions was consistent with research in the marketing discipline, and substantiated a hierarchical structure of data quality dimensions.

Using our preliminary conceptual framework, we conducted a two-phase sorting study. The first phase of the study was to sort these intermediate dimensions into a small set of categories. The second phase was to confirm that these dimensions indeed belonged to the categories in our preliminary conceptual framework.

Method

We first created four categories (see column 1 of Table 2) based on our preliminary conceptual framework, following Moore and Benbasat [31]. We then grouped the 20 intermediate dimensions into these four categories (see column 2 of Table 2). Our initial grouping was based on our understanding of these categories and dimensions. The sorting study provided the data to test this initial grouping and to make adjustments

Copyright © 2001 All Rights Reserved

in the assignment of dimensions to target categories (see column 3 of Table 2), which will be further discussed.

The Sorting Study: Phase 1

Subjects: Thirty subjects from industry were selected to participate in the overall sorting procedure. These subjects were enrolled in an evening M.B.A. class in another large university. Eighteen of these 30 subjects were randomly selected to participate in the first phase.

Design: Each of the 20 dimensions, along with a description, was printed on a 3×5-inch card, as shown in appendix D1. These cards were used by each of the subjects in the study to group the 20 dimensions into a small set of categories. In contrast to phase 2, the subjects for phase 1 were not given a prespecified set of categories each with a name and description. The study was pretested by two graduate-level MIS students to clarify any ambiguity in the design or instruction.

Procedure: The study was run by a third party who was not aware of the goal of this research, in order to avoid any bias by the authors. Before performing the actual sorting task, subjects performed a trial sort using dimensions other than these 20 dimensions to ensure that they understood the procedure. In the actual sorting task, subjects were given instructions to group the 20 cards into three to five piles. The subjects were then asked to label each of their piles.

The Sorting Study: Phase 2

The original assignment of dimensions to categories was adjusted based on the results from the phase 1 study. For example, as shown in column 3 of Table 2, *completeness* is moved from the *accuracy* category to the *relevancy* category because only four subjects assigned this dimension to the former category, whereas twelve assigned it to the latter. This was a reasonable adjustment because completeness could be interpreted within the context of the data consumer's task instead of our initial interpretation that completeness was part of the accuracy category.

In addition, five dimensions were eliminated: traceability, variety of data sources, ease of operation, flexibility, and cost-effectiveness. These dimensions were eliminated for both of the following two reasons: First, subjects did not consistently assign the dimension to any category. For example, seven subjects assigned cost-effectiveness to the relevancy category, three assigned it to the other three categories, and eight assigned it to a self-defined category. Second, the dimension was not ranked highly in terms of importance. For example, cost-effectiveness was ranked 14 out of 20.

The purpose of the second phase of the sorting study was to confirm that the dimensions indeed belonged to these adjusted categories.

Subjects: The remaining twelve subjects from our subject pool participated in this phase of study.

Design: For each category of dimensions revealed from phase 1, the authors provided a label, as shown in appendix D2, based on the underlying dimensions.

Copyright © 2001 All Rights Reserved

Descriptive phrases, rather than single words, were used as labels to avoid confounding category labels with any of the dimension labels.

Procedure: The third party that ran the phase 1 study also ran the phase 2 study. The procedure for phase 2 was similar to that of phase 1, with the exception that subjects were instructed to place each of the dimension cards into the category that best represents that dimension.

Findings

In this section, we present the results from the two-phase study. We using the adjusted target categories to tabulate the results from the phase 1 study. As shown in Table 3, the overall placement ratio of dimensions within target categories was 70 percent. This indicated that these 15 dimensions were generally being placed in the appropriate categories.

These results, together with the adjustment of dimensions within the target categories, led us to refine the four target categories as follows:

1. The extent to which data values are in conformance with the actual or true values;
2. The extent to which data are applicable (pertinent) to the task of the data user;
3. The extent to which data are presented in an intelligible and clear manner; and
4. The extent to which data are available or obtainable.

These four descriptions were used as the category labels for the phase 2 study. The results from the phase 2 study (Table 4) showed that the overall placement ratio of dimensions within target categories was 81 percent.

Toward a Hierarchical Framework of Data Quality

IN OUR SORTING STUDY, WE LABELED EACH CATEGORY on the basis of our preliminary conceptual framework and our initial grouping of the dimensions. For example, we labeled as *accuracy* the category that includes *accuracy, objectivity, believability*, and *reputation*. Similarly, we labeled the three other categories as *relevancy, representation*, and *accessibility*. We used such a labeling so that we would not introduce any additional interpretations or biases into the sorting tasks.

However, such representative labels did not necessarily capture the essence of the underlying dimensions as a group. For example, as a whole, the group of dimensions labeled accuracy was richer than that conveyed by the label accuracy. Thus, we reexamined the underlying dimensions confirmed for each of the four categories and picked a label that captured the essence of the entire category. For example, we relabeled *accuracy* as *intrinsic DQ* because the underlying dimensions captured the intrinsic aspect of data quality.

As a result of this reexamination, we relabeled two of the four categories. The resulting categories, therefore, are: *intrinsic DQ, contextual DQ, representational DQ*,

Copyright © 2001 All Rights Reserved

Table 3. Results from the Phase 1 Study (15 Dimensions and 18 Subjects)

Target categories	Actual categories: Accuracy	Relevancy	Representation	Accessibility	N/A	Total	Target (%)
Accuracy	57	10	2	1	2	72	79
Relevancy	16	56	11	2	5	90	62
Repres.	8	4	50	5	5	72	69
Access.	1	2	3	26	4	36	72

Total item placements: 270
Hits: 189
Overall hits ratio: 70%

Note: A target category is a hypothesized category based on our preliminary conceptual framework. An actual category is the category selected by the subjects for a dimension. "N/A" denotes "Not Applicable," which means that the actual category does not fit into any target category.

Table 4. Results from the Phase 2 Study (15 Dimensions and 12 Subjects)

Target categories	Actual categories: Accuracy	Relevancy	Representation	Accessibility	Total	Target (%)
Accuracy	43	3	1	0	48	90
Relevancy	7	44	3	6	60	73
Repres.	2	6	40	0	48	83
Access.	1	1	3	19	24	79

Total item placements: 180
Hits: 146
Overall hits ratio: 81%

and *accessibility DQ* (see figure 2). Intrinsic DQ denotes that data have quality in their own right. Accuracy is merely one of the four dimensions underlying this category. Contextual DQ highlights the requirement that data quality must be considered within the context of the task at hand; that is, data must be relevant, timely, complete, and appropriate in terms of amount so as to add value. Representational DQ and accessibility DQ emphasize the importance of the role of systems; that is, the system must be accessible but secure, and the system must present data in such a way that they are interpretable, easy to understand, and represented concisely and consistently.

This hierarchical framework confirms and substantiates the preliminary framework that we proposed. Below we elaborate on these four categories, relate them to the literature, and discuss some future research directions.

Copyright © 2001 All Rights Reserved

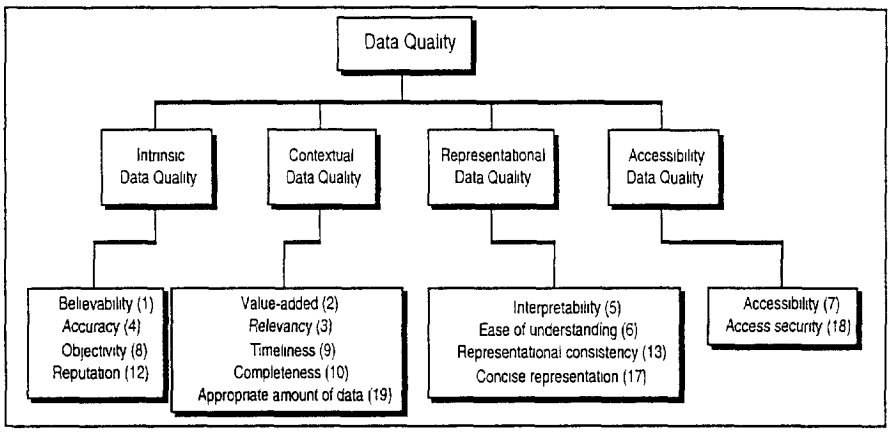


Figure 2. A Conceptual Framework of Data Quality

Intrinsic Data Quality

Intrinsic DQ includes not only accuracy and objectivity, which are evident to IS professionals, but also *believability* and *reputation*. This suggests that, contrary to the traditional development view, data consumers also view believability and reputation as an integral part of intrinsic DQ; accuracy and objectivity alone are not sufficient for data to be considered of high quality. This is analogous to some aspects of product quality. In the product quality area, dimensions of quality emphasized by consumers are broader than those emphasized by product manufacturers. Similarly, intrinsic DQ encompasses more than the accuracy and objectivity dimensions that IS professionals strive to deliver. This finding implies that IS professionals should also ensure the believability and reputation of data. Research on data source tagging [45, 48] is a step in this direction.

Contextual Data Quality

Some individual dimensions underlying contextual DQ were reported previously; for example, completeness and timeliness [4]. However, contextual DQ was not explicitly recognized in the data quality literature. Our grouping of dimensions for contextual DQ revealed that data quality must be considered within the context of the task at hand. This was consistent with the literature on graphical data representation, which concluded that the quality of a graphical representation must be assessed within the context of the data consumer's task [41].

Since tasks and their contexts vary across time and data consumers, attaining high contextual data quality is a research challenge [29, 39]. One approach is to parameterize contextual dimensions for each task so that a data consumer can specify what type of task is being performed and the appropriate contextual parameters for that task. Below we illustrate such a research prototype.

Copyright © 2001 All Rights Reserved

During Desert Storm combat operations in the Persian Gulf, naval researchers recognized the need to explicitly incorporate contextual DQ into information systems in order to deliver more timely and accurate information. As a result, a prototype is being developed that will be deployed to the U.S. aircraft carriers as stand-alone image exploitation tools [33]. This prototype parameterizes contextual dimensions for each task so that a pilot or a strike planner can specify what type of task (e.g., strike plan or damage assessment) is being performed and the appropriate contextual parameters (relevant images in terms of location, currency, resolution, and target type) for that task.

Representational Data Quality

Representational DQ includes aspects related to the format of the data (*concise* and *consistent representation*) and meaning of data (*interpretability* and *ease of understanding*). These two aspects suggest that for data consumers to conclude that data are well represented, they must not only be concise and consistently represented, but also interpretable and easy to understand.

Issues related to meaning and format arise in database systems research in which format is addressed as part of syntax, and meaning as part of semantic reconciliation. One focus of current research in that area is context interchange among heterogeneous database systems [36]. For example, currency figures in the context of a U.S. database are typically in dollars, whereas those in a Japanese database are likely to be in yen. This type of context belongs to the representational DQ, instead of contextual DQ, which deals with the data consumer's task.

Accessibility Data Quality

Information systems professionals understand accessibility DQ well. Our research findings show that data consumers also recognize its importance. Our findings appear to differ from the literature that treats accessibility as distinct from information quality (see, e.g., [9]). A closer examination reveals that accessibility is presumed (i.e., perfect accessibility DQ) in earlier information quality literature because hard-copy reports were used instead of on-line data. In contrast, data consumers in our research access computers for their information needs, and therefore, view accessibility DQ as an important data quality aspect. However, there is little difference between treating accessibility DQ as a category of overall data quality, or separating it from other categories of data quality. In either case, accessibility needs to be taken into account.

Summary and Conclusions

TO IMPROVE DATA QUALITY, WE NEED TO UNDERSTAND WHAT DATA QUALITY means to data consumers (those who use data). This research develops a hierarchical framework that captures the aspects of data quality that are important to data consumers. Specifically, 118 data quality attributes collected from data consumers are

Copyright © 2001 All Rights Reserved

consolidated into twenty dimensions, which in turn are grouped into four categories. Using this framework, information systems professionals will be able to better understand and meet their data consumers' data quality needs.

In developing this framework, we conducted a two-stage survey and a two-phase sorting study. The resulting framework has four data quality (DQ) categories: (1) intrinsic DQ consists of accuracy, objectivity, believability, and reputation; (2) contextual DQ consists of value-added, relevancy, timeliness, completeness, and appropriate amount of data; (3) representational DQ consists of interpretability, ease of understanding, representational consistency, and concise representation; and (4) accessibility DQ consists of accessibility and access security.

Intrinsic DQ denotes that data have quality in their own right. Contextual DQ highlights the requirement that data quality must be considered within the context of the task at hand. Representational DQ and accessibility DQ emphasize the importance of the role of systems. These findings are consistent with our understanding that high-quality data should be intrinsically good, contextually appropriate for the task, clearly represented, and accessible to the data consumer.

The salient feature of this research study is that quality attributes of data are collected from data consumers instead of being defined theoretically or based on researchers' experience. Furthermore, this study provides additional evidence for a hierarchical structure of data quality dimensions. At a basic level, the justification for the framework is that a data quality framework does not exist and one is needed so that data quality can be measured, analyzed, and improved in a valid way. Information systems researchers have chosen many different dependent variables for assessing information systems in general, and data quality in particular, with little empirical or theoretical foundation for their choice. This framework provides a basis for deciding which aspects of data quality to use in any research study.

The framework is further justified by the use of well-established empirical methods in its development. Thus, we argue that the framework is methodologically sound, and that it is complete from the perspective of data consumers. Furthermore, this framework will be useful as a basis for measuring, analyzing, and improving data quality. While we have only anecdotal evidence to support this claim, that anecdotal evidence is strong and convincing.

This framework was used effectively in industry and government. For example, IS managers in one investment firm thought they had perfect data quality (in terms of accuracy) in their organizational databases. However, in their discussion with data consumers using this framework, they found several deficiencies: (1) additional information about data sources was needed so that data consumers could assess the reputation and believability of data; (2) data downloaded to servers from the mainframe were not sufficiently timely for some data consumers' tasks; and (3) the currency ($, £, or ¥) and unit (thousands or millions) of financial data from different servers were implicit so data consumers could not always interpret and understand these data correctly.

Based on this hierarchical framework, several research directions can be pursued. First, a questionnaire could be developed to measure perceived data quality. The data

Copyright © 2001 All Rights Reserved

quality categories and their underlying dimensions in this framework would provide the constructs to be measured. Second, methods for improving the quality of data as perceived by data consumers could be developed. Such methods could include user training to change the quality of data as perceived by data consumers. Third, the framework could be useful as a checklist during data requirements analysis. That is, many of the data quality characteristics are actually system requirements or user training requirements. Finally, since a single empirical study is never sufficient to validate the completeness of a framework, further research is needed to apply this framework in specific work contexts.

NOTES

1. *Computerworld*, September 28, 1992, p. 80–84.
2. *The Wall Street Journal*, May 26, 1992, p. B6.
3. We refer to the characteristics of data quality as data quality attributes or as measurement items to distinguish them from data quality dimensions which result from the factor analysis throughout this section
4. Surveys with significant missing values (nine surveys) or surveys returned by academics (fourteen surveys) were not considered viable and therefore were eliminated from our analysis.

REFERENCES

1. Arnold, S.E. Information manufacturing: the road to database quality. *Database, 15*, 5 (1992), 32.
2. Bailey, J.E., and Pearson, S.W Development of a tool for measuring and analyzing computer user satisfaction. *Management Science, 29*, 5 (1983), 530–545.
3. Ballou, D.P., and Pazer, H.L Designing information systems to optimize the accuracy-timeliness tradeoff. *Information Systems Research (ISR), 6*, 1 (1995), 51–72.
4. Ballou, D.P., and Pazer, H.L Modeling data and process quality in multi-input, multi-output information systems. *Management Science, 31*, 2 (1985), 150–162
5. Ballou, D.P., and Tayi, K.G. Methodology for allocating resources for data quality enhancement *Communications of the ACM, 32*, 3 (1989), 320–329.
6. Ballou, D.P.; Wang, R.Y ; Pazer, H.; and Tayi, K.G. Modeling data manufacturing systems to determine data product quality. Total Data Quality Management (TDQM) Research Program, MIT Sloan School of Management, No. TDQM–93–09, 1993.
7. Bodnar, G. Reliability modeling of internal control systems. *Accounting Review, 50*, 4 (1975), 747–757.
8. Churchill, G.A. *Marketing Research: Methodological Foundations*, 5th ed. Chicago: Dryden Press, 1991
9. Culnan, M. The dimensions of accessibility to online information: implications for implementing office information systems. *ACM Transactions on Office Information Systems, 2*, 2 (1984), 141–150
10. Cureton, E.E., and D'Agostino, R.B. *Factor Analysis: An Applied Approach*. Hillsdale, NJ: Lawrence Erlbaum, 1983.
11. Cushing, B.E. A mathematical approach to the analysis and design of internal control systems. *Accounting Review, 49*, 1 (1974), 24–41.
12. Delone, W.H., and McLean, E.R. Information systems success: the quest for the dependent variable. *Information Systems Research, 3*, 1 (1992), 60–95.
13. Deming, E.W. *Out of the Crisis* Cambridge, MA: Center for Advanced Engineering Study, MIT, 1986.
14. Deshpande, R. The organizational context of market research use. *Journal of Marketing, 46*, 4 (1992), 92–101.

Copyright © 2001 All Rights Reserved

15. Dobyns, L., and Crawford-Mason, C. *Quality or Else: The Revolution in World Business.* Boston: Houghton Mifflin, 1991.

16. Emery, J.C. *Organizational Planning and Control Systems: Theory and Technology.* New York: Macmillan, 1969.

17. Griffin, A., and Hauser, J.R. The voice of the customer. *Marketing Science, 12*, 1 (1993), 1–27.

18. Hauser, J.R., and Clausing, D. The house of quality. *Harvard Business Review, 66*, 3 (1988), 63–73.

19. Huh, Y.U.; Keller, F.R.; Redman, T C.; and Watkins, A.R. Data quality. *Information and Software Technology, 32*, 8 (1990), 559–565.

20. Ives, B.; Olson, M.H.; and Baroudi, J.J. The measurement of user information satisfaction. *Communications of the ACM, 26*, 10 (1983), 785–793.

21. Johnson, J.R.; Leitch, R.A.; and Neter, J Characteristics of errors in accounts receivable and inventory audits. *Accounting Review, 56*, 2 (1981), 270–293.

22. Juran, J.M. *Juran on Leadership for Quality An Executive Handbook.* New York: The Free Press, 1989.

23. Juran, J.M., and Gryna, F.M. *Quality Planning and Analysis*, 2d ed. New York: McGraw-Hill, 1980.

24. Knechel, W.R. A simulation model for evaluating accounting systems reliability. *Auditing. A Journal of Theory and Practice, 4*, 2 (1985), 38–62.

25. Knechel, W.R. The use of quantitative models in the review and evaluation of internal control: a survey and review. *Journal of Accounting Literature, 2* (1983), 205–219.

26. Kotler, P *Marketing Essentials* Englewood Cliffs, NJ: Prentice-Hall, 1984.

27. Kriebel, C.H. Evaluating the quality of information systems. In N. Szysperski and E. Grochla (eds.), *Design and Implementation of Computer Based Information Systems.* Germantown, PA: Sijthtoff and Noordhoff, 1979.

28. Laudon, K.C. Data quality and due process in large interorganizational record systems. *Communications of the ACM, 29*, 1 (1986), 4–11

29. Madnick, S. Integrating information from global systems: dealing with the "on- and off-ramps" of the information superhighway. *Journal of Organizational Computing, 5*, 2 (1995), 69–82.

30. Melone, N. A theoretical assessment of the user-satisfaction construct in information systems research. *Management Science, 36*, 1 (1990), 598–613.

31. Moore, G.C., and Benbasat, I. Development of an instrument to measure the perceptions of adopting an information technology innovation. *Information Systems Research, 2*, 3 (1991), 192–222.

32. Morey, R.C. Estimating and improving the quality of information in the MIS. *Communications of the ACM, 25*, 5 (1982), 337–342.

33. Page, W., and Kaomea, P. Using quality attributes to produce optimal tactical information. In *Proceedings of the Fourth Annual Workshop on Information Technologies and Systems (WITS).* Vancouver, British Columbia, Canada: 1994, pp. 145–154.

34. Redman, T.C. *Data Quality: Management and Technology.* New York: Bantam Books, 1992.

35. Ronen, B., and Spiegler, I. Information as inventory: a new conceptual view. *Information and Management, 21*, (1991), 239–247.

36. Sciore, E.; Siegel, M ; and Rosenthal, A. Using semantic values to facilitate interoperability among heterogeneous information systems. *ACM Transactions on Database Systems, 19*, 2 (1994), 254–290

37. Stevens, J. Applied multivariate statistics for the social science. Hillsdale, NJ: Lawrence Erlbaum, 1986.

38. Strong, D.M. Decision support for exception handling and quality control in office operations. *Decision Support Systems, 8*, 3 (1992), 217–227.

39. Strong, D.M.; Lee, Y.W.; and Wang, R.Y. Data quality in context. Forthcoming, 1996.

40. Strong, D.M., and Miller, S.M. Exceptions and exception handling in computerized information processes. *ACM Transactions on Information Systems, 13*, 2 (1995), 206–233.

41. Tan, J.K., and Benbasat, I. Processing of graphical information: a decomposition taxonomy to match data extraction tasks and graphical representations. *Information Systems Research, 1*, 4 (1990), 416–439

Copyright © 2001 All Rights Reserved

42. Wand, Y., and Wang, R.Y. Anchoring data quality dimensions in ontological foundations. *Communications of the ACM*, forthcoming, 1996.

43. Wang, R.Y., and Kon, H.B. Towards total data quality management (TDQM). In R.Y. Wang (ed.), *Information Technology in Action: Trends and Perspectives*. Englewood Cliffs, NJ: Prentice-Hall, 1993.

44. Wang, R.Y.; Kon, H.B.; and Madnick, S.E. Data quality requirements analysis and modeling. In *Proceedings of the 9th International Conference on Data Engineering*, Vienna, 1993, pp. 670–677.

45. Wang, R.Y.; Reddy, M.P.; and Kon, H.B. Toward quality data: an attribute-based approach. *Decision Support Systems (DSS), 13*, 1995 (1995), 349–372.

46. Wang, R.Y.; Storey, V.C ; and Firth, C.P. A framework for analysis of data quality research. *IEEE Transactions on Knowledge and Data Engineering, 7*, 4 (1995), 623–640.

47. Wang, R.Y.; Strong, D.M.; and Guarascio, L.M. An empirical investigation of data quality dimensions: a data consumer's perspective. Total Data Quality Management (TDQM) Research Program, MIT Sloan School of Management, no. TDQM–93–12, 1993

48. Wang, Y.R., and Madnick, S.E. A polygen model for heterogeneous database systems: the source tagging perspective. In *Proceedings of the 16th International Conference on Very Large Data Bases (VLDB)*, Brisbane, Australia, 1990, pp. 519–538.

49. Yu, S., and Neter, J. A stochastic model of the internal control system. *Journal of Accounting Research, 1*, 3 (1973), 273–295.

APPENDIX A: First Data Quality Survey Questionnaire

Side One

Position Prior to Attending the University (circle one): Finance Marketing Operations Personnel IT Other

Industry you worked in the previous job:

When you think of data quality, what attributes other than timeliness, accuracy, availability, and interpretability come to mind? Please list as many as possible!

PLEASE FILL OUT THIS SIDE BEFORE TURNING OVER. THANK YOU!!

Side Two

The following is a list of attributes developed for data quality:

Completeness	Flexibility	Adaptability	Reliability
Relevance	Reputation	Compatibility	Ease of Use
Ease of Update	Ease of Maintenance	Format	Cost
Integrity	Breadth	Depth	Correctness
Well-documented	Habit	Variety	Content
Dependability	Manipulability	Preciseness	Redundancy
Ease of Access	Convenience	Accessibility	Data Exchange
Understandable	Credibility	Importance	Critical

After reviewing this list, do any other attributes come to mind?

THANK YOU!

Copyright © 2001 All Rights Reserved

APPENDIX B: Second Data Quality Survey Questionnaire

Thank you for participating in this study. All responses will be held in strictest confidence.

Industry:

Job Title:

Department: Finance Marketing/Sales Operations Human Resources Accounting Information Systems Planning Other

The following is a list of adjectives and phrases which describes corporate data. When answering the questions, please think about the internal data such as sales, production, financial, and employee data that you work with or use to make decisions in your job.

We apologize for the tedious nature of the survey. Although the questions may seem repetitive, your response to each question is critical to the success of the study. Please give us the first response that comes to mind and try to use the FULL scale range available.

Section I: How important is it to you that your data are:

	Extremely important				Important				Not important at all
Accurate	1	2	3	4	5	6	7	8	9
Believable	1	2	3	4	5	6	7	8	9
Complete	1	2	3	4	5	6	7	8	9
Concise	1	2	3	4	5	6	7	8	9
Verifiable	1	2	3	4	5	6	7	8	9
Well-documented	1	2	3	4	5	6	7	8	9
Understandable	1	2	3	4	5	6	7	8	9
Well-presented	1	2	3	4	5	6	7	8	9
Up-to-date	1	2	3	4	5	6	7	8	9
Accessible	1	2	3	4	5	6	7	8	9
Adaptable	1	2	3	4	5	6	7	8	9
Aesthetically Pleasing	1	2	3	4	5	6	7	8	9
Compactly Represented	1	2	3	4	5	6	7	8	9
Important	1	2	3	4	5	6	7	8	9
Consistently Formatted	1	2	3	4	5	6	7	8	9
Dependable	1	2	3	4	5	6	7	8	9
Retrievable	1	2	3	4	5	6	7	8	9
Manipulable	1	2	3	4	5	6	7	8	9
Objective	1	2	3	4	5	6	7	8	9
Usable	1	2	3	4	5	6	7	8	9
Well-organized	1	2	3	4	5	6	7	8	9
Transportable/Portable	1	2	3	4	5	6	7	8	9
Unambiguous	1	2	3	4	5	6	7	8	9
Correct	1	2	3	4	5	6	7	8	9
Relevant	1	2	3	4	5	6	7	8	9
Flexible	1	2	3	4	5	6	7	8	9
Flawless	1	2	3	4	5	6	7	8	9
Comprehensive	1	2	3	4	5	6	7	8	9
Consistently Represented	1	2	3	4	5	6	7	8	9
Interesting	1	2	3	4	5	6	7	8	9

Copyright © 2001 All Rights Reserved

Unbiased	1	2	3	4	5	6	7	8	9
Familiar	1	2	3	4	5	6	7	8	9
Interpretable	1	2	3	4	5	6	7	8	9
Applicable	1	2	3	4	5	6	7	8	9
Robust	1	2	3	4	5	6	7	8	9
Available	1	2	3	4	5	6	7	8	9
Revealing	1	2	3	4	5	6	7	8	9
Reviewable	1	2	3	4	5	6	7	8	9
Expandable	1	2	3	4	5	6	7	8	9
Time Independent	1	2	3	4	5	6	7	8	9
Error-free	1	2	3	4	5	6	7	8	9
Efficient	1	2	3	4	5	6	7	8	9
User-friendly	1	2	3	4	5	6	7	8	9
Specific	1	2	3	4	5	6	7	8	9
Well-formatted	1	2	3	4	5	6	7	8	9
Reliable	1	2	3	4	5	6	7	8	9
Convenient	1	2	3	4	5	6	7	8	9
Extendable	1	2	3	4	5	6	7	8	9
Critical	1	2	3	4	5	6	7	8	9
Well-defined	1	2	3	4	5	6	7	8	9
Reusable	1	2	3	4	5	6	7	8	9
Clear	1	2	3	4	5	6	7	8	9
Cost-effective	1	2	3	4	5	6	7	8	9
Auditable	1	2	3	4	5	6	7	8	9
Precise	1	2	3	4	5	6	7	8	9
Readable	1	2	3	4	5	6	7	8	9

Section II: How important is it to you that your data can be:

	Extremely important				Important			Not important at all	
Easily Aggregated	1	2	3	4	5	6	7	8	9
Easily Accessed	1	2	3	4	5	6	7	8	9
Easily Compared to Past Data	1	2	3	4	5	6	7	8	9
Easily Changed	1	2	3	4	5	6	7	8	9
Easily Questioned	1	2	3	4	5	6	7	8	9
Easily Downloaded/Uploaded	1	2	3	4	5	6	7	8	9
Easily Joined with Other Data	1	2	3	4	5	6	7	8	9
Easily Updated	1	2	3	4	5	6	7	8	9
Easily Understood	1	2	3	4	5	6	7	8	9
Easily Maintained	1	2	3	4	5	6	7	8	9
Easily Retrieved	1	2	3	4	5	6	7	8	9
Easily Customized	1	2	3	4	5	6	7	8	9
Easily Reproduced	1	2	3	4	5	6	7	8	9
Easily Traced	1	2	3	4	5	6	7	8	9
Easily Sorted	1	2	3	4	5	6	7	8	9

Section III: How important are the following to you?

	Extremely important				Important			Not important at all	
Data are certified error-free	1	2	3	4	5	6	7	8	9
Data improve efficiency	1	2	3	4	5	6	7	8	9
Data give you a competitive edge	1	2	3	4	5	6	7	8	9

Copyright © 2001 All Rights Reserved

Data cannot be accessed by competitors	1	2	3	4	5	6	7	8	9
Data contain adequate detail	1	2	3	4	5	6	7	8	9
Data are in finalized form	1	2	3	4	5	6	7	8	9
Data contain no redundancy	1	2	3	4	5	6	7	8	9
Data are of proprietary nature	1	2	3	4	5	6	7	8	9
Data can be personalized	1	2	3	4	5	6	7	8	9
Data are not easily corrupted	1	2	3	4	5	6	7	8	9
Data meet all of your requirements	1	2	3	4	5	6	7	8	9
Data add value to your operations	1	2	3	4	5	6	7	8	9
Data are continuously collected	1	2	3	4	5	6	7	8	9
Data continuously presented in same format	1	2	3	4	5	6	7	8	9
Data are compatible with previous data	1	2	3	4	5	6	7	8	9
Data are not overwhelming	1	2	3	4	5	6	7	8	9
Data can be easily integrated	1	2	3	4	5	6	7	8	9
Data can be used for multiple purposes	1	2	3	4	5	6	7	8	9
Data are secure	1	2	3	4	5	6	7	8	9

Section IV: How important are the following to you?

	Extremely important					Important			Not important at all
The source of the data is clear	1	2	3	4	5	6	7	8	9
Errors can be easily identified	1	2	3	4	5	6	7	8	9
The cost of data collection	1	2	3	4	5	6	7	8	9
The cost of data accuracy	1	2	3	4	5	6	7	8	9
The form of presentation	1	2	3	4	5	6	7	8	9
The format of the data	1	2	3	4	5	6	7	8	9
The scope of information contained in data	1	2	3	4	5	6	7	8	9
The depth of information contained in data	1	2	3	4	5	6	7	8	9
The breadth of information contained in data	1	2	3	4	5	6	7	8	9

Copyright © 2001 All Rights Reserved

Quality of resolution	1	2	3	4	5	6	7	8	9
The storage medium	1	2	3	4	5	6	7	8	9
The reputation of the data source	1	2	3	4	5	6	7	8	9
The reputation of the data	1	2	3	4	5	6	7	8	9
The age of the data	1	2	3	4	5	6	7	8	9
The amount of data	1	2	3	4	5	6	7	8	9
You have used the data before	1	2	3	4	5	6	7	8	9
Someone has clear responsibility for data	1	2	3	4	5	6	7	8	9
The data entry process is self-correcting	1	2	3	4	5	6	7	8	9
The speed of access to data	1	2	3	4	5	6	7	8	9
The speed of operations performed on data	1	2	3	4	5	6	7	8	9
The amount and type of storage required	1	2	3	4	5	6	7	8	9
You have little extraneous data present	1	2	3	4	5	6	7	8	9
You have a variety of data and data sources	1	2	3	4	5	6	7	8	9
You have optimal data for your purpose	1	2	3	4	5	6	7	8	9
The integrity of the data	1	2	3	4	5	6	7	8	9
It is easy to tell if the data are updated	1	2	3	4	5	6	7	8	9

APPENDIX C: Descriptive Statistics for Attributes

Attribute	No. of cases	Mean	S.D.	Min	Max
Accurate	350	1.771	1.135	1	7
Believable	348	2.707	1.927	1	9
Complete	349	3.229	1.814	1	9
Concise	348	3.994	2.016	1	9
Verifiable	348	3.224	1.854	1	9
Well-documented	349	4.123	2.087	1	9
Understandable	349	2.668	1.671	1	9
Well-presented	350	3.937	2.124	1	9
Up-to-date	350	2.963	1.732	1	9
Accessible	349	3.370	1.899	1	9
Adaptable	344	4.942	2.042	1	9
Aesthetically Pleasing	350	6.589	2.085	1	9
Compactly Represented	349	5.123	2.181	1	9
Important	335	3.824	2.138	1	9
Consistently Formatted	347	4.594	2.141	1	9
Dependable	349	2.648	1.615	1	9
Retrievable	350	3.660	1.999	1	9
Manipulable	349	4.327	2.162	1	9
Objective	345	3.551	1.963	1	9

Copyright © 2001 All Rights Reserved

Data give you a competitive edge	348	3.178	2.277	1	9
Data cannot be accessed by competitors	347	4.450	2.760	1	9
Data contain adequate detail	348	3.057	1.378	1	8
Data are in finalized form	348	5.575	2.201	1	9
Data contain no redundancy	344	6.279	2.026	1	9
Data are of proprietary nature	346	5.867	2.612	1	9
Data can be personalized	345	5.759	2.390	1	9
Data are not easily corrupted	344	3.741	2.162	1	9
Data meet all of your requirements	348	3 664	2.123	1	9
Data add value to your operations	349	2.479	1.708	1	9
Data are continuously collected	347	4.608	2.443	1	9
Data continuously presented in same format	346	4.627	2.232	1	9
Data are compatible with previous data	348	3.578	1.893	1	9
Data are not overwhelming	347	4.037	2.306	1	9
Data can be easily integrated	348	4.086	1.896	1	9
Data can be used for multiple purposes	347	4.565	2.304	1	9
Data are secure	349	4.456	2.432	1	9
The source of the data is clear	350	3.291	1.836	1	9
Errors can be easily identified	347	3.089	1.584	1	8
The cost of data collection	349	4.304	2.180	1	9
The cost of data accuracy	348	4.261	2.169	1	9
The form of presentation	349	4.794	1.994	1	9
The format of the data	348	4.917	2.045	1	9
The scope of information contained in data	345	3.838	1.726	1	9
The depth of information contained in data	345	3.922	1.835	1	9
The breadth of information contained in data	344	3.872	1.796	1	9
Quality of resolution	329	5.024	1.995	1	9
The storage medium	348	6.534	2.148	1	9
The reputation of the data source	348	4.144	2.172	1	9
The reputation of the data	347	3.925	2.133	1	9
The age of the data	350	3.640	2.044	1	9
The amount of data	347	5.009	2.125	1	9
You have used the data before	345	6.107	2.228	1	9
Someone has clear responsibility for data	347	3.744	2.271	1	9
The data entry process is self-correcting	344	4.695	2.362	1	9

Copyright © 2001 All Rights Reserved

The speed of access to data	347	3.934	1.992	1	9
The speed of operations performed on data	348	4.687	2.194	1	9
The amount and type of storage required	349	6.209	2.030	1	9
You have little extraneous data present	345	5.797	2.003	1	9
You have a variety of data and data sources	344	4.712	2.234	1	9
You have optimal data for your purpose	345	3.554	2.126	1	9
The integrity of the data	345	2.371	1 571	1	9
It is easy to tell if the data are updated	348	3 609	1 926	1	9
Easy to exchange data with others	346	4.945	2.311	1	9
Access to data can be restricted	347	4.988	2.514	1	9

APPENDIX D: The Two-Phase Sorting Study

D1: Instruction and Content for Phase 1

Instruction 1

Group the 20 data quality dimensions into several categories (between 3 and 5) where the dimensions within each category in your opinion represent similar attributes of high-quality data. (Note: A data quality dimension may also be isolated into its own category if you see fit to do so.)

Example 3 × 5 Card

BELIEVABILITY

The extent to which data are accepted or regarded as true, real, and credible.

(1)

Content of the Remaining Nineteen 3 × 5 Dimension Cards

2. VALUE-ADDED: The extent to which data are beneficial and provide advantages from their use.
3. RELEVANCY: The extent to which data are applicable and helpful for the task at hand.
4. ACCURACY: The extent to which data are correct, reliable, and certified free of error.
5. INTERPRETABILITY: The extent to which data are in appropriate language and units and the data definitions are clear.

Copyright © 2001 All Rights Reserved

6. EASE OF UNDERSTANDING: The extent to which data are clear without ambiguity and easily comprehended.
7. ACCESSIBILITY: The extent to which data are available or easily and quickly retrievable.
8. OBJECTIVITY: The extent to which data are unbiased (unprejudiced) and impartial.
9. TIMELINESS: The extent to which the age of the data is appropriate for the task at hand.
10. COMPLETENESS: The extent to which data are of sufficient breadth, depth, and scope for the task at hand.
11. TRACEABILITY: The extent to which data are well documented, verifiable, and easily attributed to a source.
12. REPUTATION: The extent to which data are trusted or highly regarded in terms of their source or content.
13. REPRESENTATIONAL CONSISTENCY: The extent to which data are always presented in the same format and are compatible with previous data.
14. COST-EFFECTIVENESS: The extent to which the cost of collecting appropriate data is reasonable.
15. EASE OF OPERATION: The extent to which data are easily managed and manipulated (i.e., updated, moved, aggregated, reproduced, customized).
16. VARIETY OF DATA AND DATA SOURCES: The extent to which data are available from several differing data sources.
17. CONCISE: The extent to which data are compactly represented without being overwhelming (i.e., brief in presentation, yet complete and to the point).
18. ACCESS SECURITY: The extent to which access to data can be restricted and hence kept secure.
19. APPROPRIATE AMOUNT OF DATA: The extent to which the quantity or volume of available data is appropriate.
20. FLEXIBILITY: The extent to which data are expandable, adaptable, and easily applied to other needs.

Instruction 2

Label the categories that you have created with an overall definition (word or two/three-word phrase) that best describes/summarizes the data quality dimensions within each category.

D2: Instruction and Content for Phase 2

Instruction

Group each of the data quality dimensions into one of the following four categories. In case of conflict, choose the best-fitting category for the dimension. All dimensions must be categorized.

Copyright © 2001 All Rights Reserved

Content of the Four 3 × 5 Category Cards

Category 1: The extent to which data values are in conformance with the actual or true values.

Category 2: The extent to which data are applicable to or pertain to the task of the data user.

Category 3: The extent to which data are presented in an intelligible and clear manner

Category 4: The extent to which data are available or obtainable.

Copyright © 2001 All Rights Reserved

Copyright © 2001 All Rights Reserved